RailSync

Smart Railway Block & Permit Coordination

A cloud-based coordination platform for Engineering, OHE, S&T and Traffic teams.

Cloud Computing | Prototype for ECR, Danapur Division

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Live prototype: railsync-green.vercel.app

The Problem

- **Operating & Traffic** (Control & Stations - G&SR)
- **Electrical & Traction** (OHE / TRD / 25kV Catenary)
- **Civil Engineering** (Permanent Way / P-Way & Bridges)
- **Signal & Telecommunication** (S&T / Interlocking)
- **Mechanical** (Carriage & Wagon / C&W & Rolling Stock)
- **Safety & Security** (RPF / Safety Directorate)



No shared block status & approval workflow

Railway maintenance blocks need coordinated approvals across several departments, but fragmented workflows make request status, dependencies and overlapping-block conflicts harder to manage.

Fragmented approvals

Block requests, power isolation, traffic feasibility and S&T notices run through separate workflows.

Risk of overlapping blocks

Without a shared conflict-checking mechanism, overlapping requests are harder to detect and coordinate.

Limited audit visibility

Scattered records make it harder to reconstruct who approved what, and when.

Example: an OHE maintenance request may need power isolation, traffic feasibility, interlocking checks and field clearance before work can begin.

The Solution

One corridor. One approval chain. One tamper-resistant record.

RailSync digitises block requests, permits and clearances, from the first request to the final reconnection memo.

RailSync coordinates and records decisions; authorized railway personnel retain operational authority.

Digital requisition

KM geofencing, time window and impact toggles such as 25 kV power isolation.

7-state lifecycle

Draft to Normalized, with Redis fencing tokens to help prevent overlapping digital grants.

Two-person human approvals

Each authorized role completes its required checks and sign-off before the workflow advances.

Tamper-resistant audit trail

Time-stamped approvals and event history in a protected ledger for traceability.

Offline-aware field app: observations can sync later; safety-critical approvals need an active connection.

How It Works ?

1 Drafting Request

Engineering or OHE submits KM range, time window and impact.

2 Request Submitted

TPC coordinates and records the authorized power-isolation step.

3 Dept. Overview

Section Controller checks the block against G&SR rules.

4 Awaiting Approval

Station Master confirms interlocking and point clamping.

5 Approved

Notify teams of signalling, track-bonding and shared-infrastructure dependencies.

6 Action on Track

Authorized field staff verify conditions and carry out the prescribed physical earthing.

7 Normalized

Worksite cleared, joint safety clearance, then authorized restoration and record.

Audit Trail

Designed to record every workflow event in a protected audit ledger (SHA-256 hash chain, S3 Object Lock).

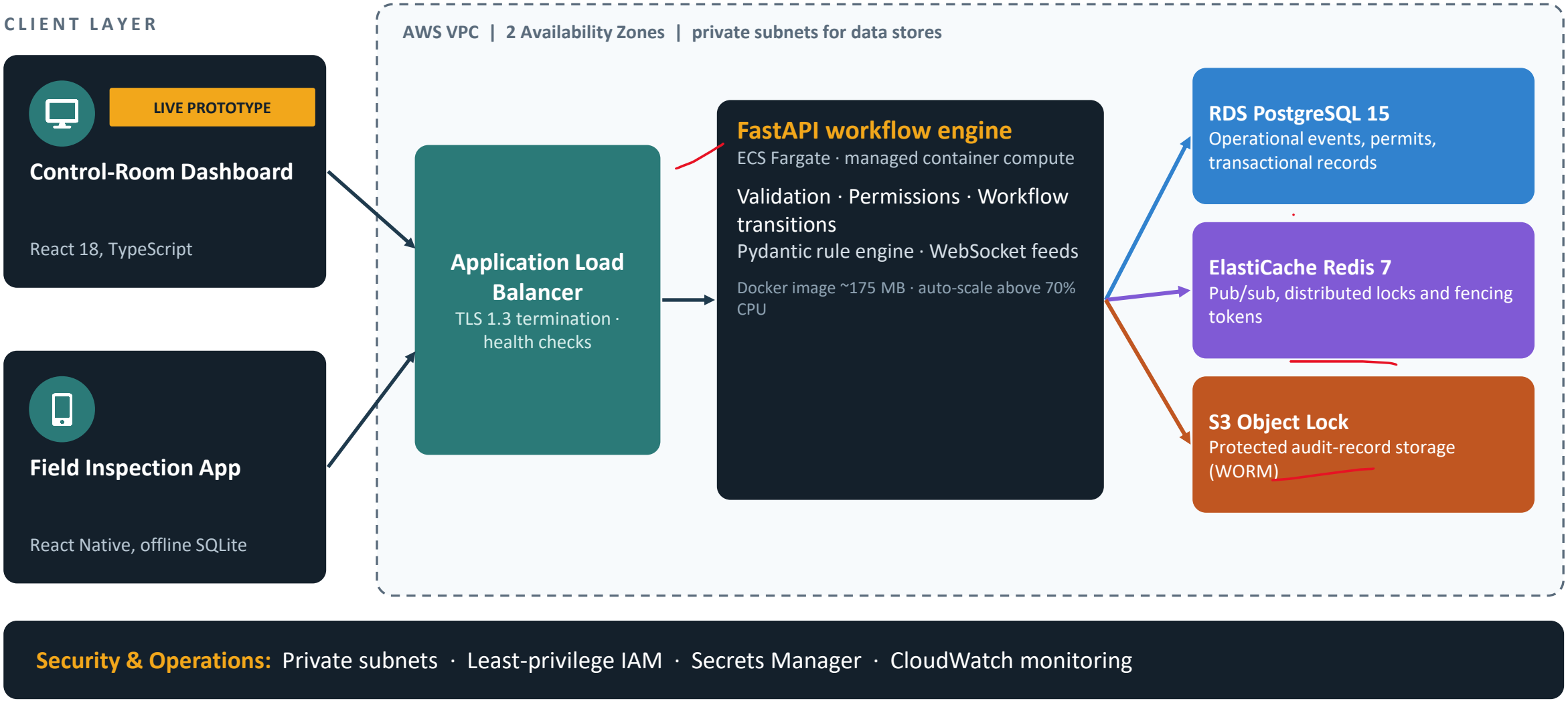
After work: Completion → Joint safety verification → Authorized restoration → Audit record

Each transition needs the authorized role's sign-off. This slide shows what people do; slide 6 shows the system states behind it.

Cloud Architecture

PROPOSED ARCHITECTURE

Amber badge = built in the live prototype. Everything else is proposed.



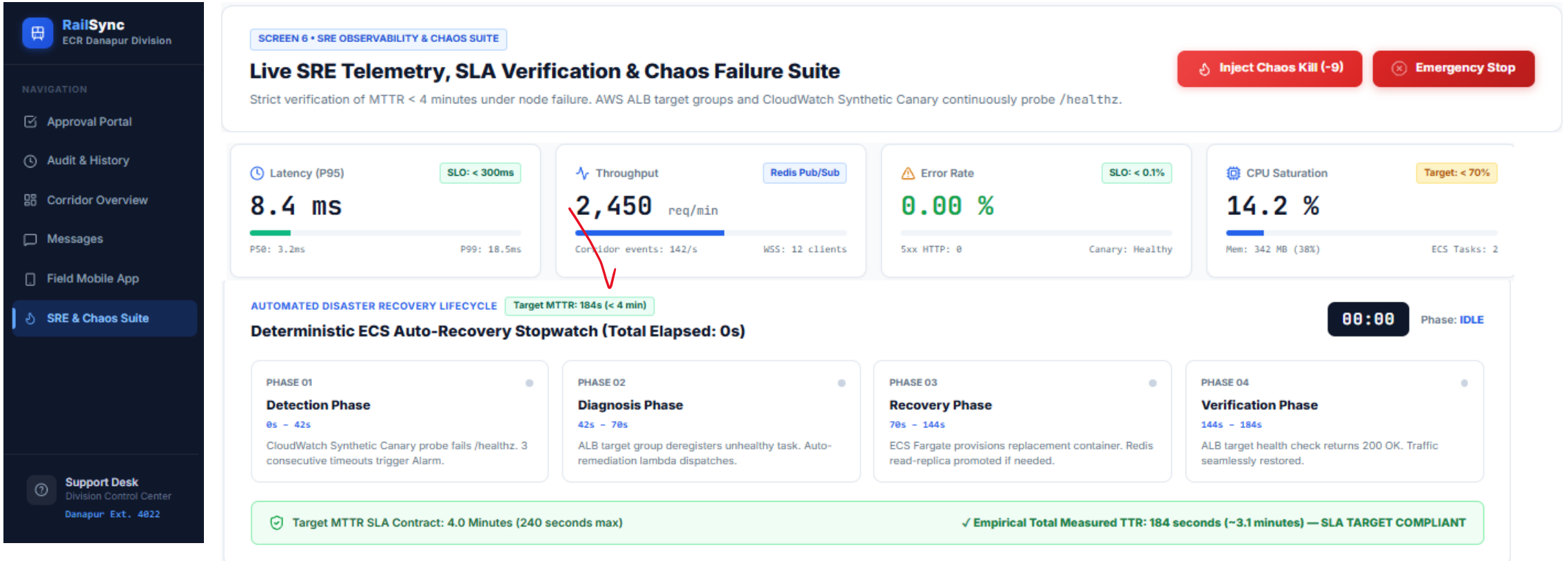
Security & Operations: Private subnets · Least-privilege IAM · Secrets Manager · CloudWatch monitoring

Built-in Safety Engine

7-STATE LIFECYCLE

LIFECYCLE STATE MACHINE · 7 STATES

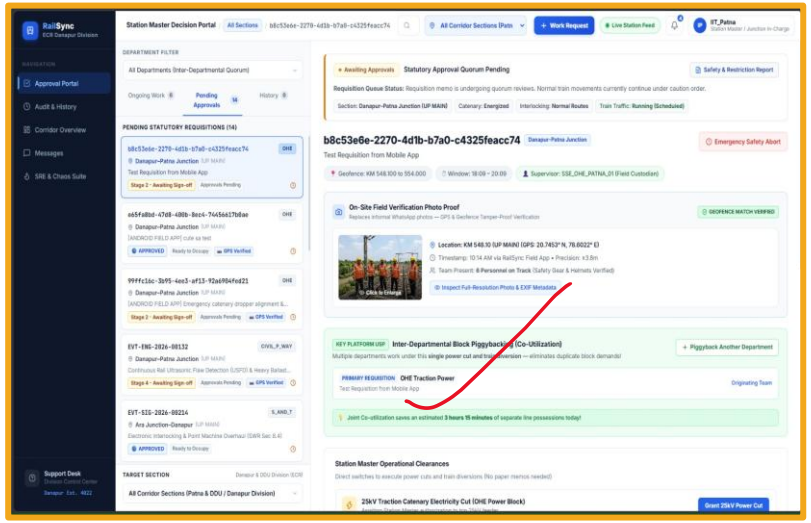
1 Draft	2 Submitted	3 Dep. Resolved	4 Awaiting Approval	5 Approved	6 Active on Track	7 Normalized
Requisition created	Sent into the approval workflow	Department dependencies and conflicts resolved	Authorised roles review and sign off	All required sign-offs recorded	Work under permit; block is live	Track fit certificate signed, 25 kV power restored



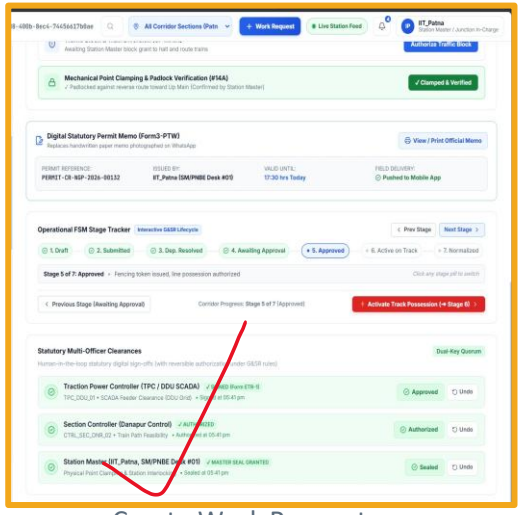
Live Prototype

railsync-green.vercel.app

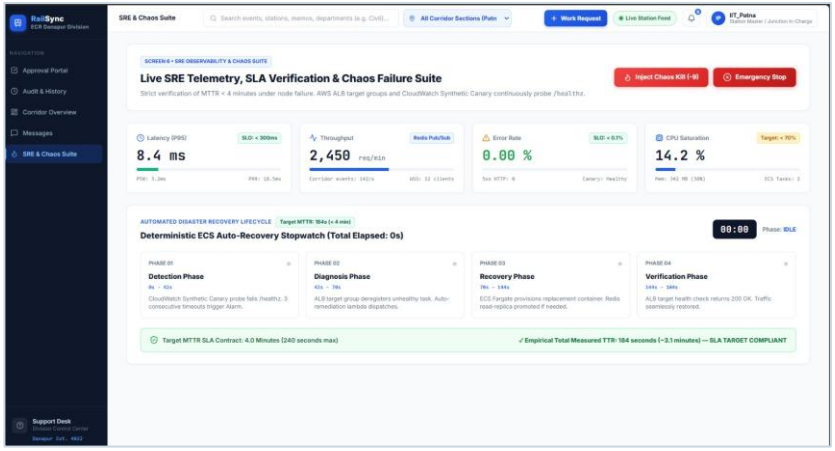
WEB PORTAL · Station Master / control room



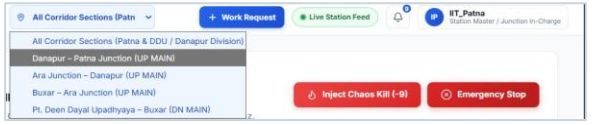
Station Master Decision portal



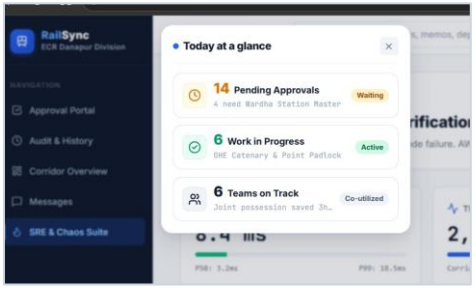
Create Work Request



SRE & Chaos Suite

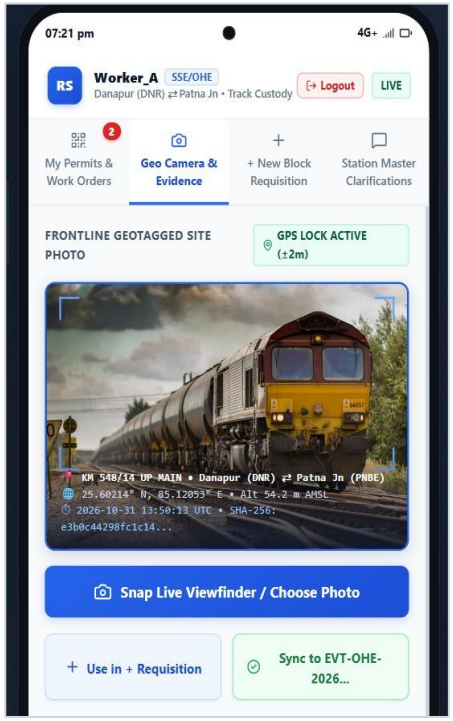


Corridor section filter

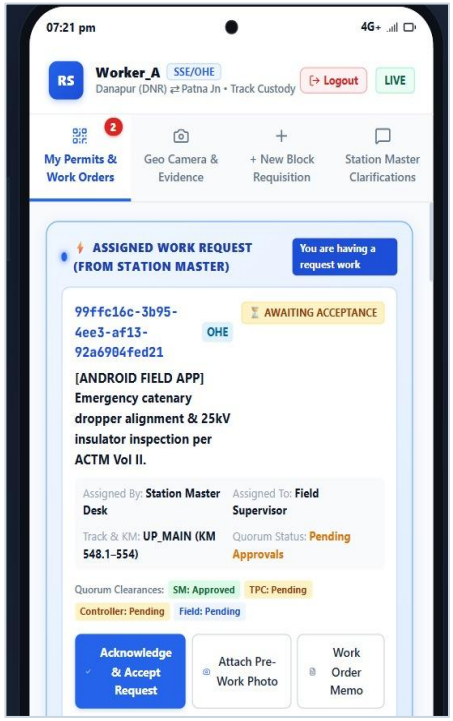


“Today at a glance” summary

FIELD MOBILE APP · filled-up pages



Geo camera & evidence



Permits & work orders

https://field-app-railsync.vercel.app

Fits Alongside Existing Systems

A coordination layer on top of today's railway procedures, not a rip-and-replace.

IT DOESN'T REPLACE EXISTING SYSTEMS

Statutory procedures stay

G&SR rules, ETR forms and permit-to-work remain in force. RailSync digitises them.

Humans keep authority

Authorised railway staff make every operational decision; software records and checks.

Existing tools keep running

Control-office and train-running tools continue unchanged; integration is a later step.

HOW IT SAVES TIME

Approvals in one chain

Departments see one shared status instead of chasing separate memos and calls.

Conflicts caught early

Overlapping blocks are flagged by the checks before a block is approved.

Audit without digging

Who approved what, and when, is one tamper-resistant record, not scattered papers.

FUTURE POSSIBILITIES

Predictive conflict detection

ML-based analysis to flag likely clashing block windows.

Live GIS track map

Kilometre overlays and section status on a map.

Integration and alerts

APIs to existing tools, SMS/push escalation, e-signatures.

RailSync coordinates and records; it does not replace statutory railway procedures or authorised human decisions.

Technical Feasibility

Frontend	React 18, TypeScript, Tailwind; React Native
Backend	Python 3.11, FastAPI, Pydantic v2
Data	PostgreSQL 15 Multi-AZ, Redis 7
Compute	Docker, ECS Fargate, load balancer
Security	Private subnets, IAM, Secrets Manager
Testing / Audit	tests/ suite, load testing, failure injection; S3 Object Lock, SHA-256 chain

TARGET

99.9%

Monthly availability

EXPECTED

185 ms

Expected P95 latency · target < 300 ms

EXPECTED

3.2 min

Expected recovery time · limit < 5 min

ESTIMATED

$145 / mo

AWS cost, proposed workload



Who It Serves & What's Next

TARGET AUDIENCE

Section Controllers
Traffic feasibility and block coordination

Traction Power Controllers (TPC)
Power-isolation coordination

Station Masters
Interlocking and point-clamping confirmation

S&T Teams
Signalling-related notices and dependencies

Track Engineers / OHE Supervisors
Field work and permit-to-work coordination

Divisional Authorities (DOM/DRM)
Escalation and oversight

PROPOSED ENHANCEMENTS

Predictive conflict detection
Proposed ML-based analysis to flag potentially conflicting block windows.

Live GIS track map
Planned map view with kilometre overlays and section status.

System integration
Planned APIs to existing control-office and train-running tools.

Alerts and e-signatures
Planned SMS or push escalation and digital sign-offs.

Roadmap: Pilot → Integration → Scale-up

Designed for better coordination, clearer approvals and traceable maintenance records.